Lab 9: Complexity Families + Color

Visualization Methods in Data Analysis by R

Group 4, Variant 5

2026-03-09

Contents

Part 1: Complexity Families (Munzner) . . . 1
Part 2: Color Spaces . . . 3
Part 3: Color Palette Types . . . 5
Part 4: Color Accessibility . . . 8
HCL Palette Browser . . . 11
Conclusions . . . 14

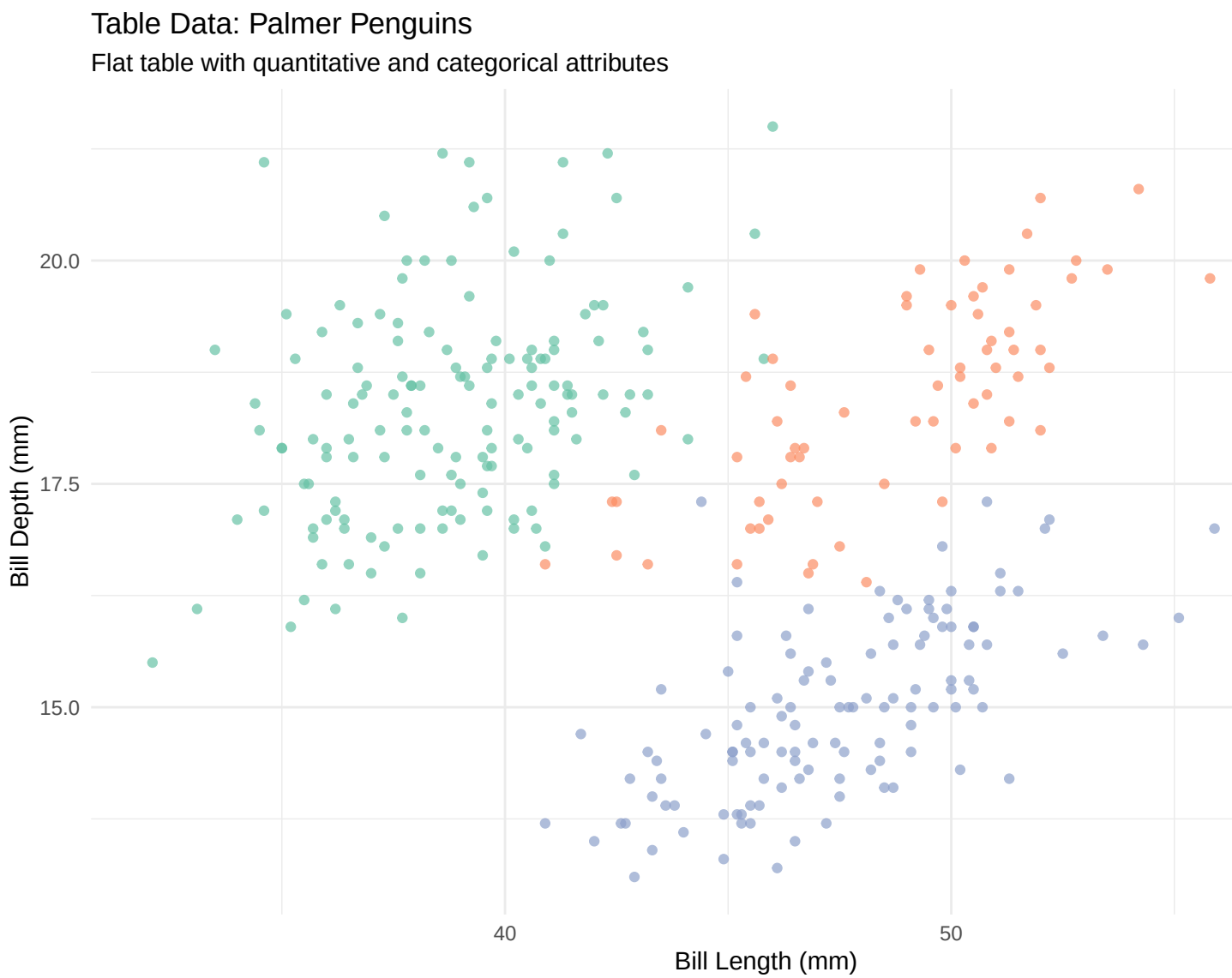
Part 1: Complexity Families (Munzner)

Dataset types

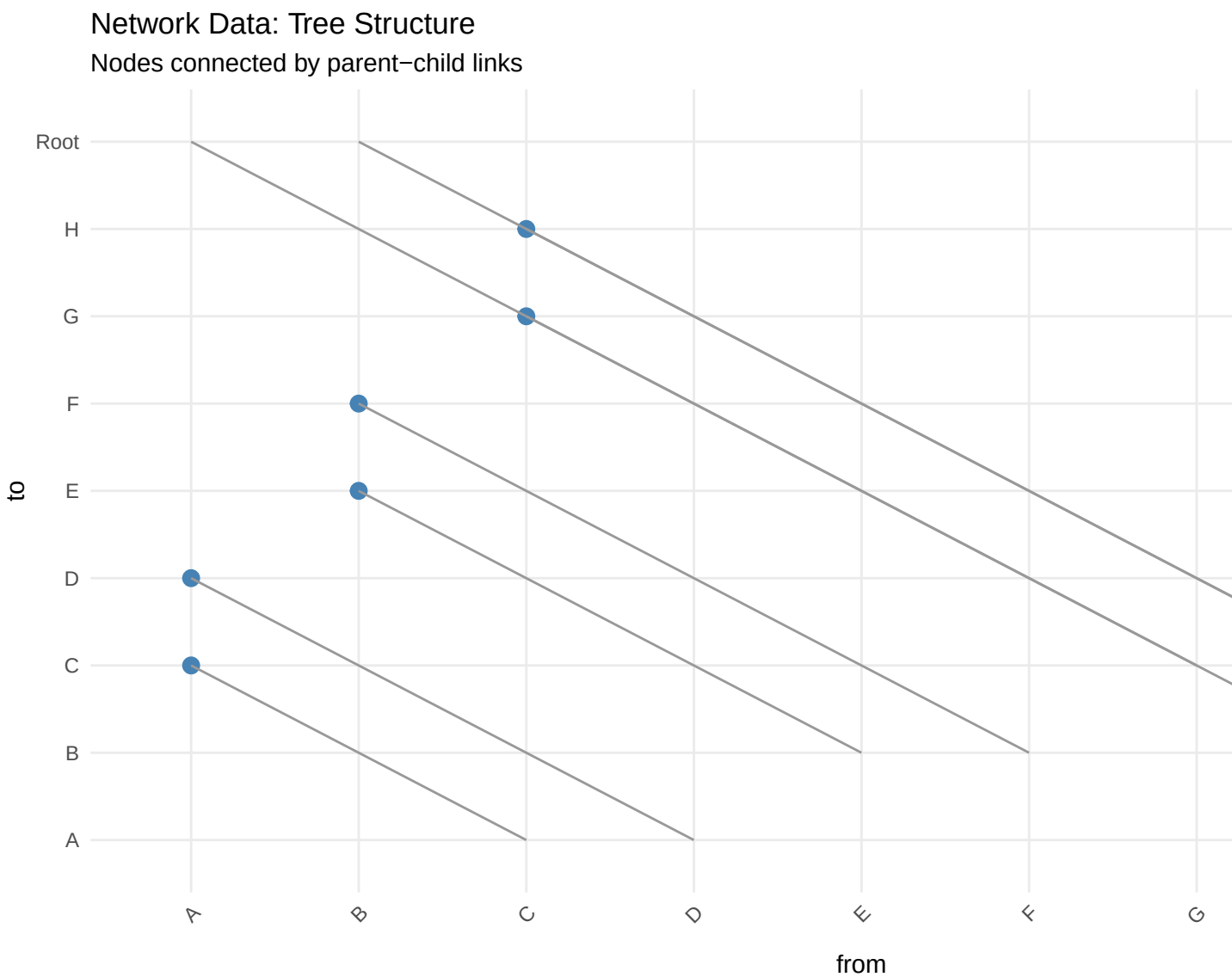
Family	Examples	Key Structure
Tables	Flat table, multidimensional table	Rows = items, columns = attributes
Networks	Trees, general graphs	Nodes + links
Spatial	Fields (continuous), geometry	Spatial position is data
Sets	Lists, clusters, partitions	Membership relationships

Classifying datasets

```r
library(palmerpenguins)
data(penguins)
penguins_clean <- penguins |> drop_na()

# TABLE example: Palmer Penguins
ggplot(penguins_clean, aes(bill_length_mm, bill_depth_mm, color = species)) +
  geom_point(alpha = 0.7) +
  scale_color_brewer(palette = "Set2") +
  labs(title = "Table Data: Palmer Penguins",
       subtitle = "Flat table with quantitative and categorical attributes",
       x = "Bill Length (mm)", y = "Bill Depth (mm)") +
  theme_minimal()
```




```r
# NETWORK example: simple hierarchy (simulated)
tree_df <- tibble(
  from = c("Root", "Root", "A", "A", "B", "B", "C", "C"),
  to = c("A", "B", "C", "D", "E", "F", "G", "H"),
  level = c(1, 1, 2, 2, 2, 2, 3, 3)
)

ggplot(tree_df, aes(x = from, y = to)) +
  geom_point(size = 3, color = "steelblue") +
  geom_segment(aes(xend = to, yend = from), color = "gray60") +
  labs(title = "Network Data: Tree Structure",
       subtitle = "Nodes connected by parent-child links") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

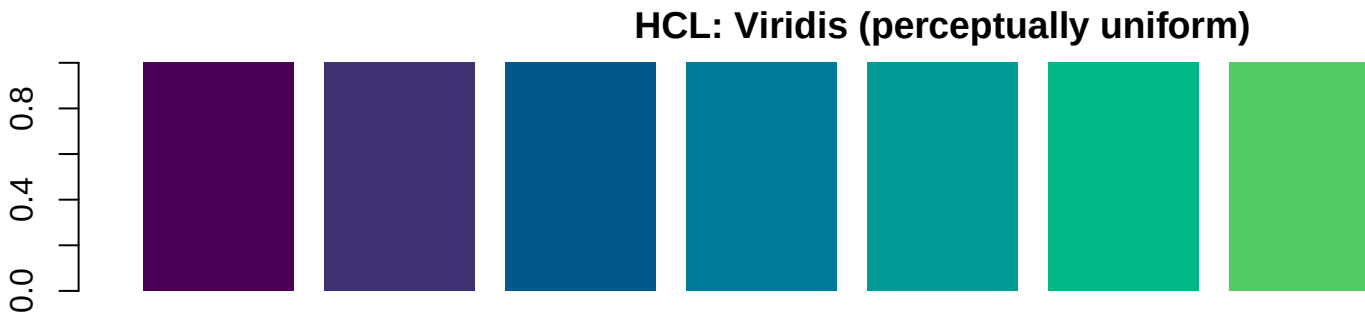


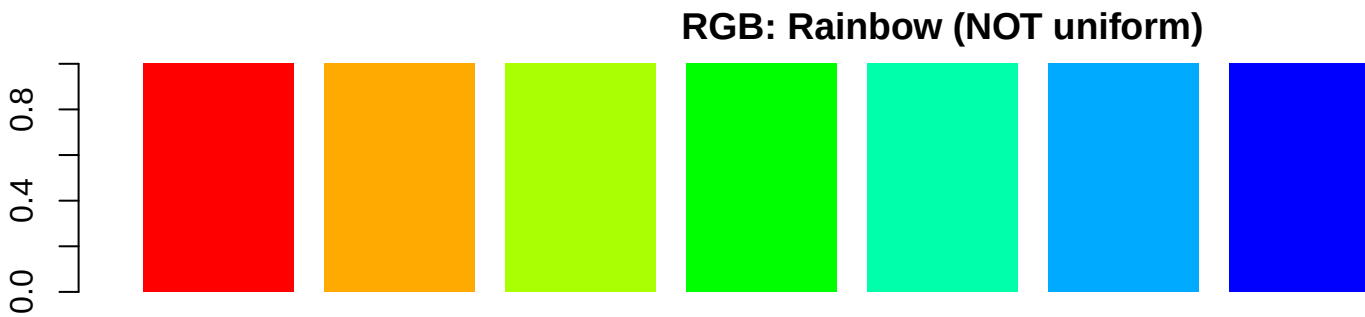
Part 2: Color Spaces

RGB vs HCL

- **RGB**: Device-oriented, not perceptually uniform
- **HCL**: Hue-Chroma-Luminance, perceptually uniform (preferred)

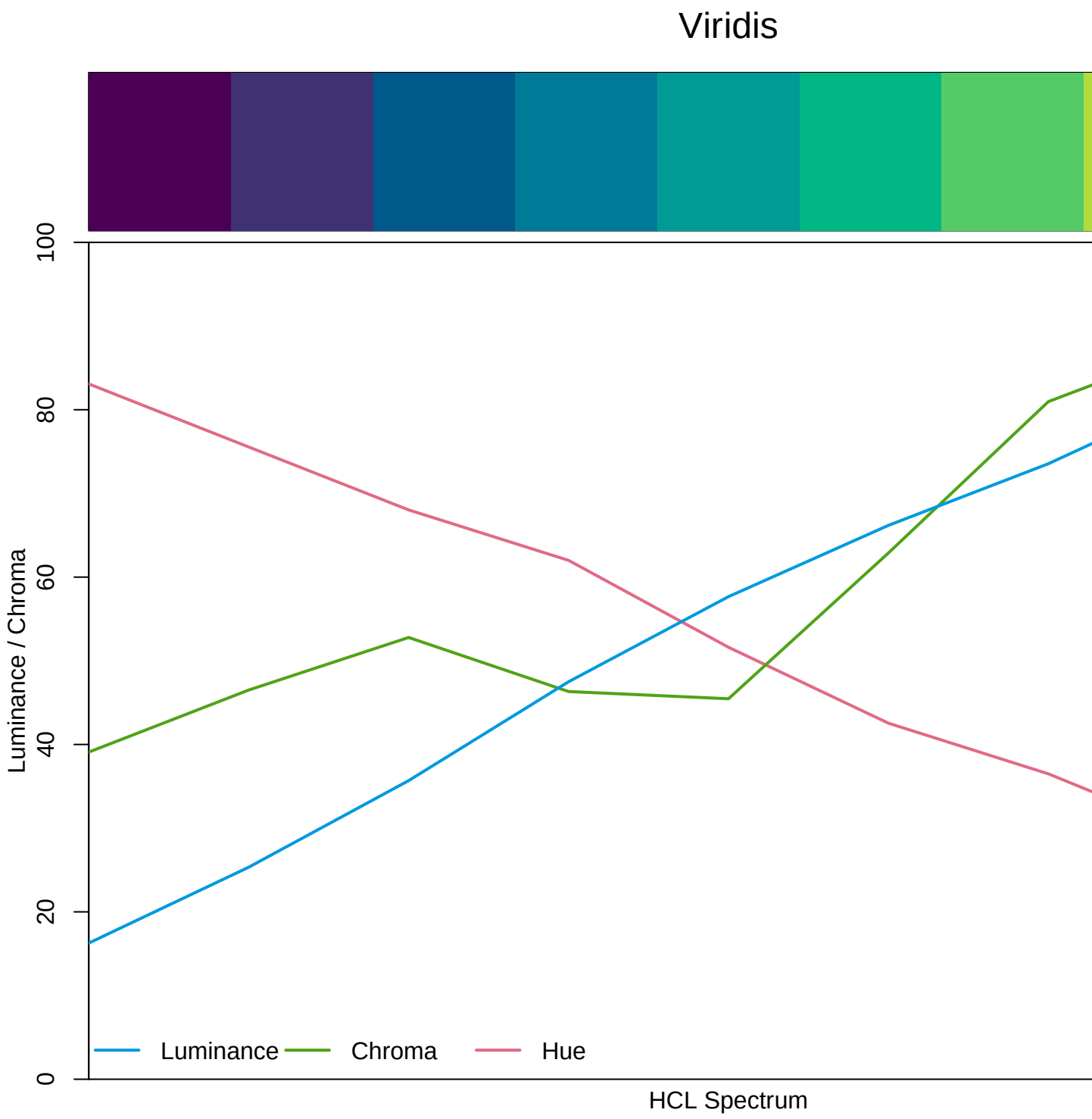
```
# Compare color spaces
hcl_pal <- hcl.colors(9, "viridis")
rainbow_pal <- rainbow(9)

par(mfrow = c(2, 1), mar = c(2, 4, 2, 1))
barplot(rep(1, 9), col = hcl_pal, border = NA, main = "HCL: Viridis (perceptually uniform)")
barplot(rep(1, 9), col = rainbow_pal, border = NA, main = "RGB: Rainbow (NOT uniform)")
```





Spectral properties

```
specplot(hcl.colors(9, "viridis"), main = "Viridis")
```

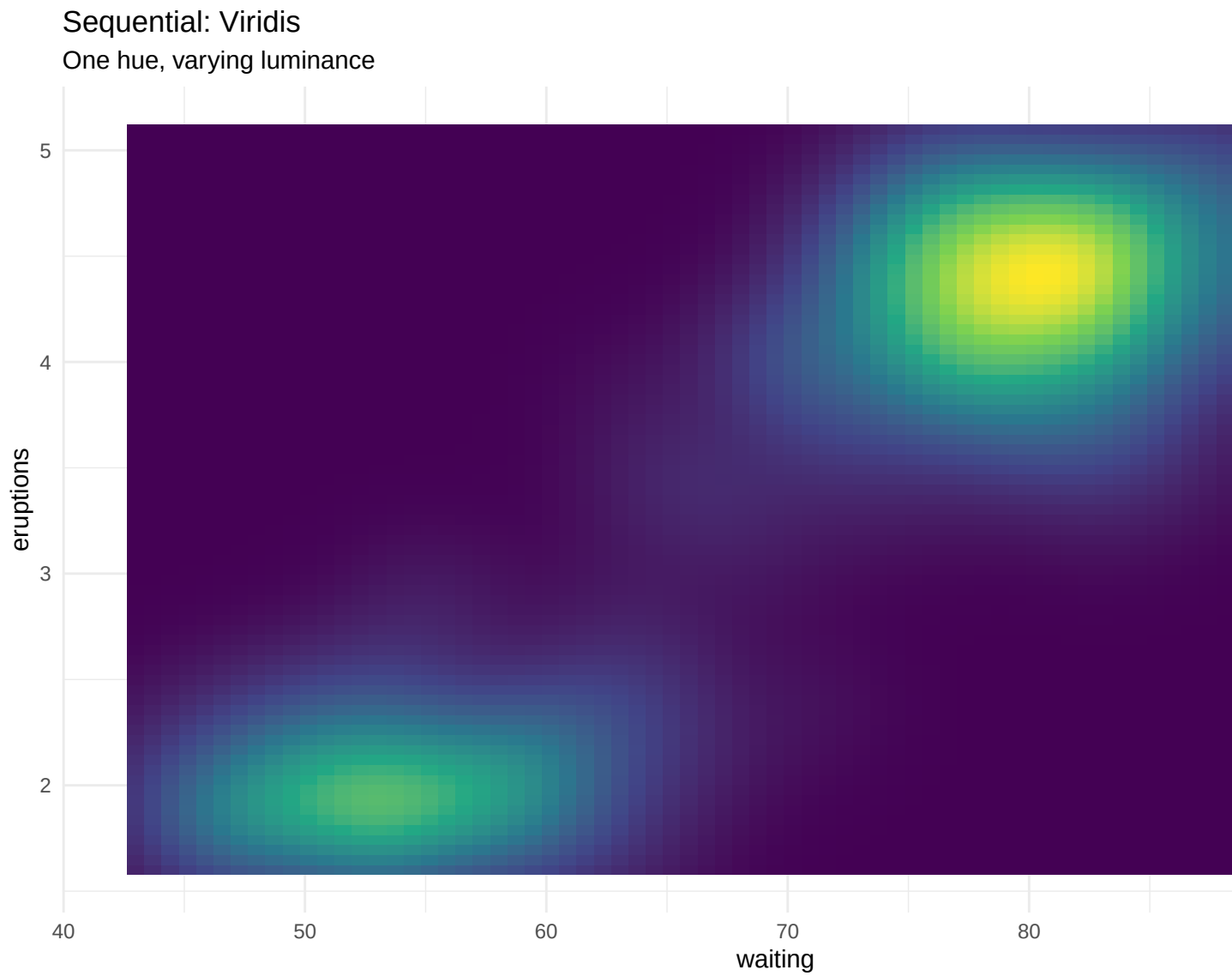


Part 3: Color Palette Types

Sequential palettes

For ordered data (low to high):

```r
ggplot(faithfuld, aes(waiting, eruptions, fill = density)) +
  geom_tile() +
  scale_fill_viridis_c(option = "viridis") +
  labs(title = "Sequential: Viridis",
       subtitle = "One hue, varying luminance") +
  theme_minimal()
```

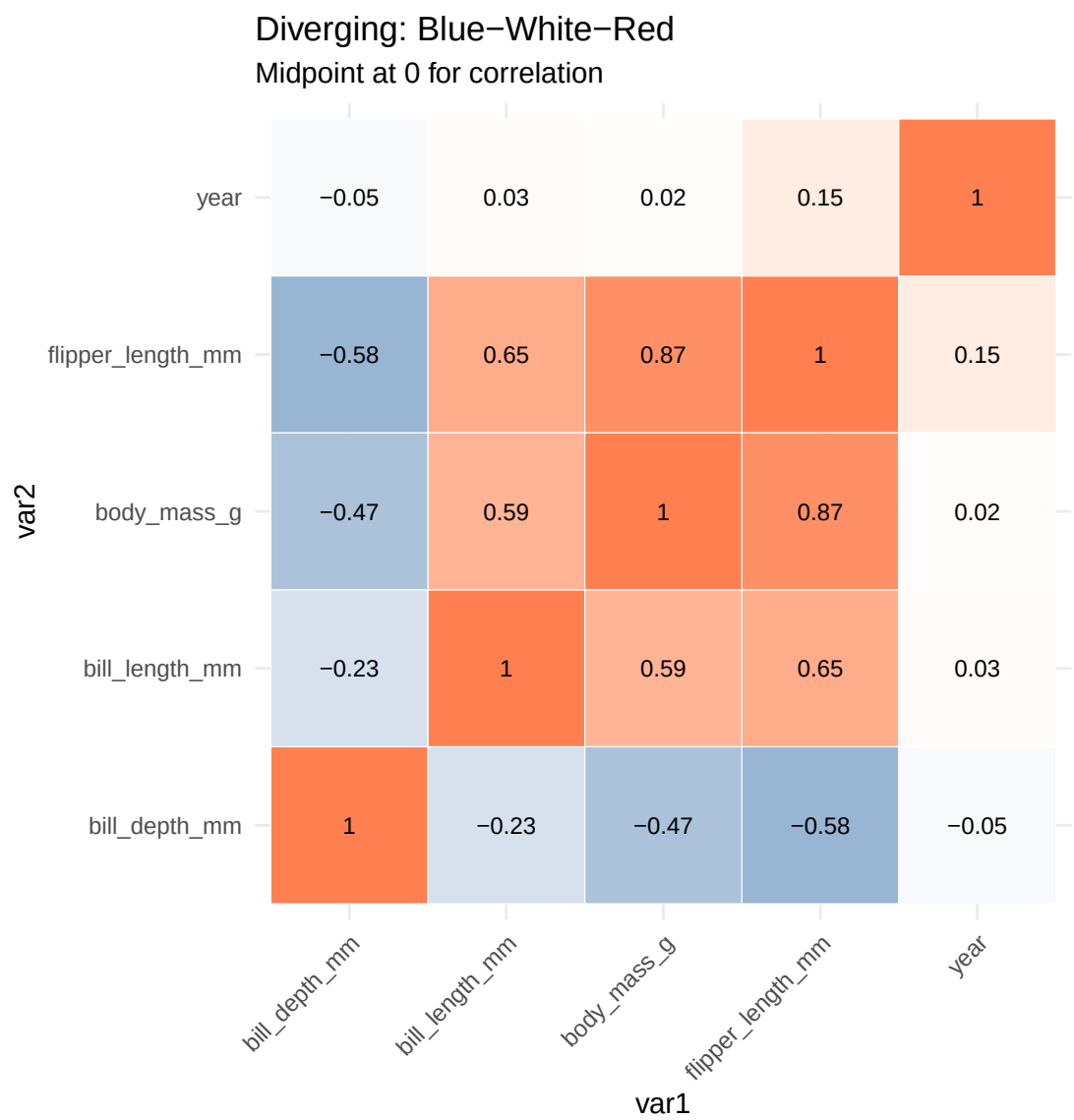


Diverging palettes

For data with meaningful midpoint:

```r
cor_matrix <- penguins_clean |>
  select(where(is.numeric)) |>
  cor()

cor_long <- cor_matrix |>
```

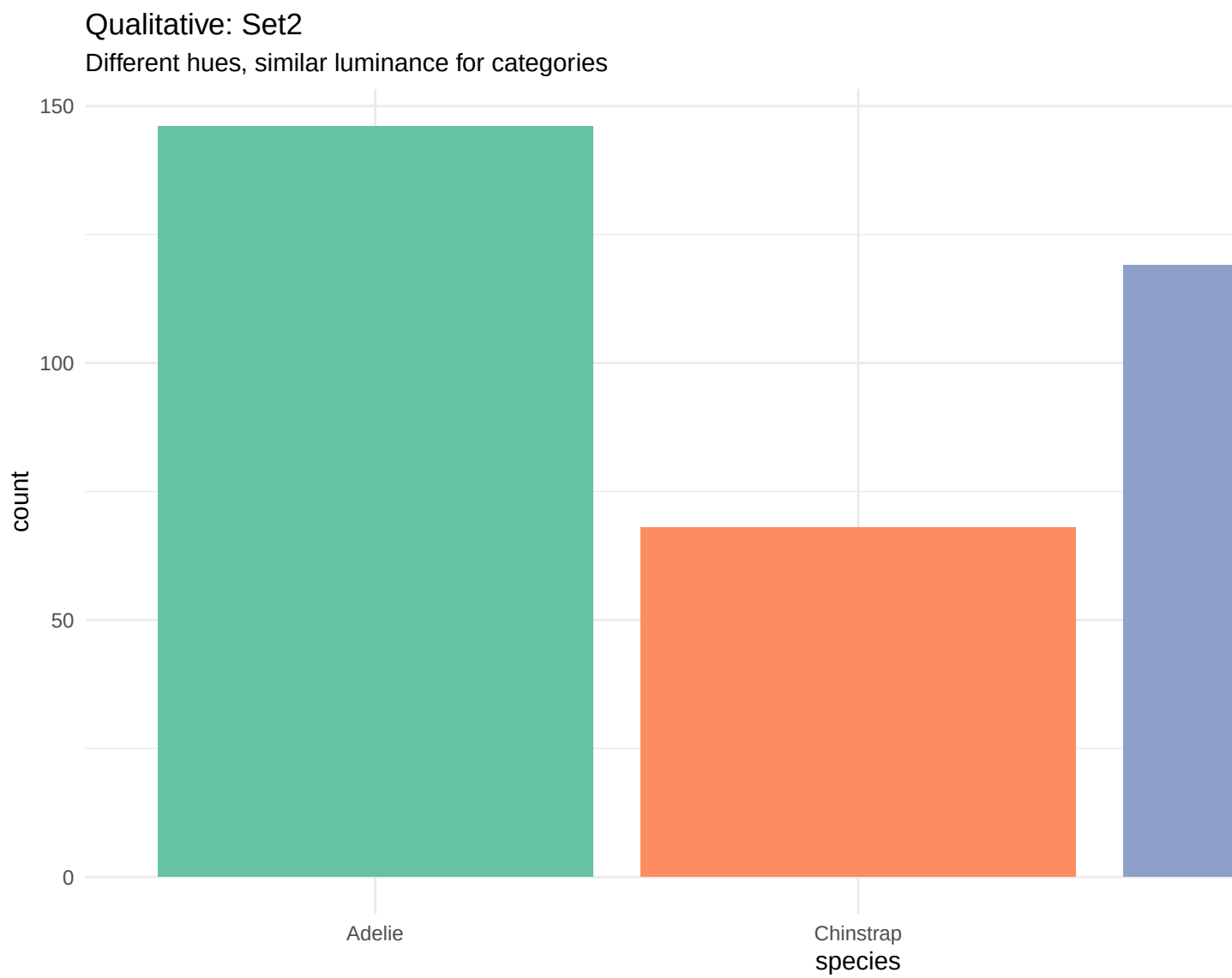
```r
  as.data.frame() |>
  rownames_to_column("var1") |>
  pivot_longer(-var1, names_to = "var2", values_to = "r")

ggplot(cor_long, aes(var1, var2, fill = r)) +
  geom_tile(color = "white") +
  geom_text(aes(label = round(r, 2)), size = 3) +
  scale_fill_gradient2(low = "steelblue", mid = "white", high = "coral",
                       midpoint = 0) +
  labs(title = "Diverging: Blue-White-Red",
       subtitle = "Midpoint at 0 for correlation") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  coord_fixed()
```



Qualitative palettes

For nominal categories:

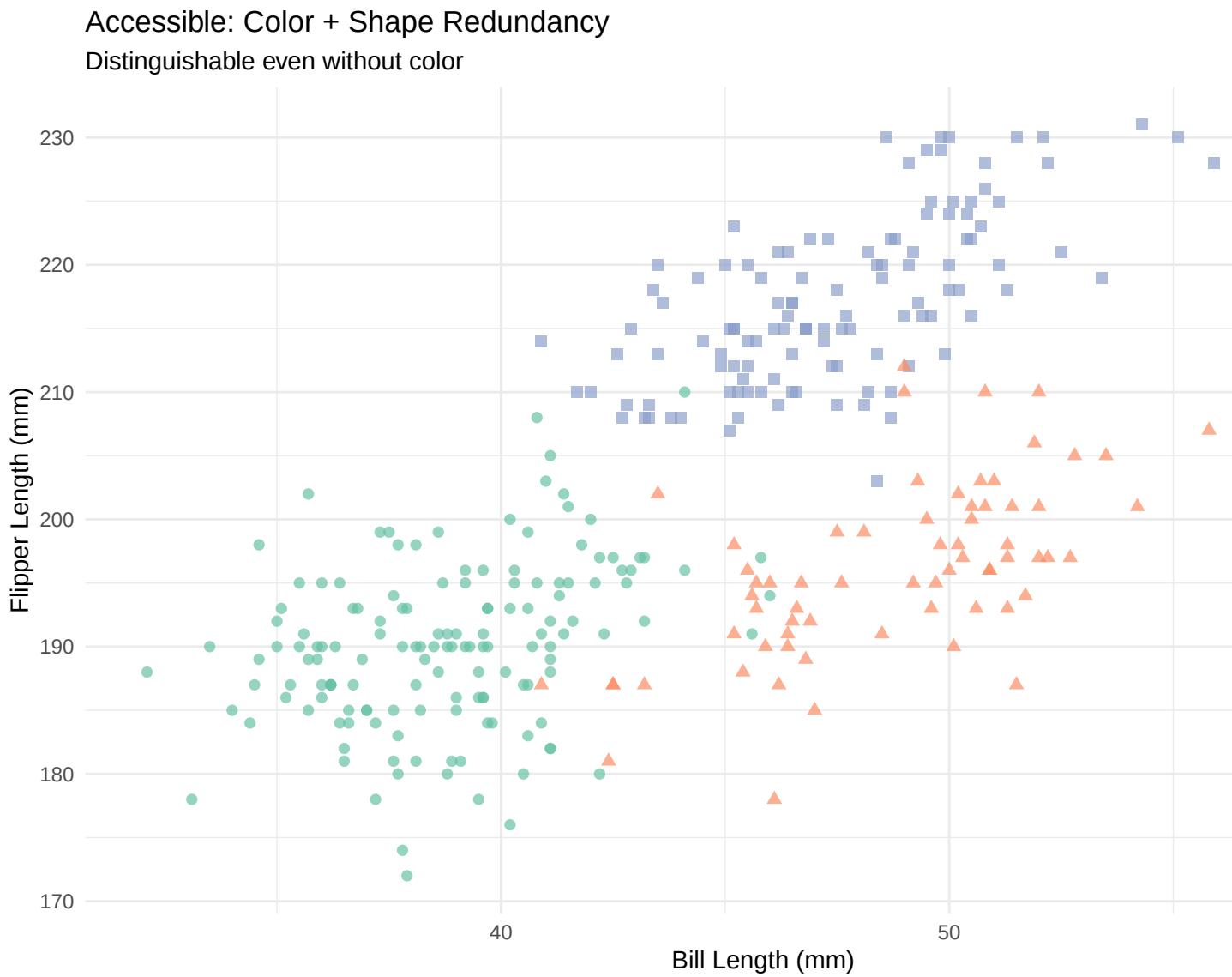
```
ggplot(penguins_clean, aes(species, fill = species)) +
  geom_bar(show.legend = FALSE) +
  scale_fill_brewer(palette = "Set2") +
  labs(title = "Qualitative: Set2",
       subtitle = "Different hues, similar luminance for categories") +
  theme_minimal()
```



Part 4: Color Accessibility

- ~8% of males and ~0.5% of females have color vision deficiency
- Always test with colorblind simulators
- Avoid red-green as sole distinguishing colors
- Use redundant encodings (shape + color)

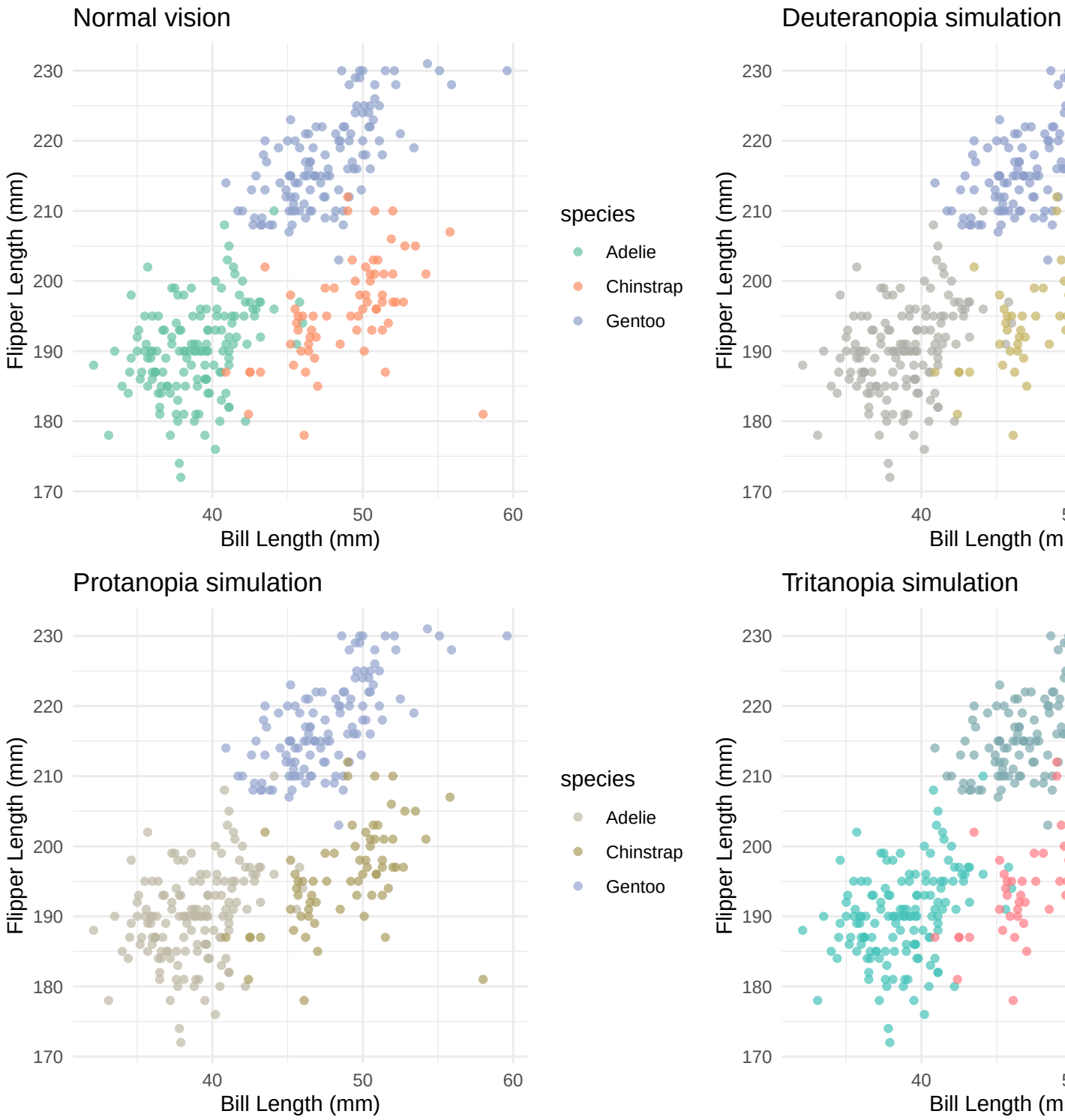
```r
# Good practice: redundant encoding
ggplot(penguins_clean, aes(bill_length_mm, flipper_length_mm,
                           color = species, shape = species)) +
  geom_point(size = 2, alpha = 0.7) +
  scale_color_brewer(palette = "Set2") +
  labs(title = "Accessible: Color + Shape Redundancy",
       subtitle = "Distinguishable even without color",
       x = "Bill Length (mm)", y = "Flipper Length (mm)") +
  theme_minimal()
```



Simulating color blindness

```r
base_colors <- RColorBrewer::brewer.pal(3, "Set2")

p <- ggplot(penguins_clean, aes(bill_length_mm, flipper_length_mm, color = species)) +
```

```r
  geom_point(alpha = 0.7) +
  labs(x = "Bill Length (mm)", y = "Flipper Length (mm)") +
  theme_minimal()

p_normal <- p + scale_color_manual(values = base_colors) +
  labs(title = "Normal vision")
p_deutan <- p + scale_color_manual(values = deutan(base_colors)) +
  labs(title = "Deuteranopia simulation")
p_protan <- p + scale_color_manual(values = protan(base_colors)) +
  labs(title = "Protanopia simulation")
p_tritan <- p + scale_color_manual(values = tritan(base_colors)) +
  labs(title = "Tritanopia simulation")

(p_normal + p_deutan) / (p_protan + p_tritan)
```



HCL Palette Browser

```r
hcl_palettes(plot = TRUE, n = 7)
```

Qualitative

Pastel 1
Dark 2
Dark 3
Set 2
Set 3
Warm
Cold
Harmonic
Dynamic

Sequential (single−hue)

Grays
Light Grays
Blues 2
Blues 3
Purples 2
Purples 3
Reds 2
Reds 3
Greens 2
Greens 3
Oslo

Sequential (multi−hue)

Purple−Blue
Red−Purple
Red−Blue
Purple−Orange
Purple−Yellow
Blue−Yellow
Green−Yellow
Red−Yellow
Heat
Heat 2
Terrain
Terrain 2
Viridis
Plasma
Inferno
Rocket
Mako
Dark Mint
Mint
BluGrn
Teal
TealGrn
Emrld
BluYl
ag_GrnYl
Peach
PinkYl
Burg
BurgYl
RedOr
OrYel
Purp
PurpOr
Sunset
Magenta
SunsetDark
ag_Sunset
BrwnYl
YlOrRd
YlOrBr
OrRd
Oranges
YlGn
YlGnBu
Reds
RdPu
PuRd
Purples
PuBuGn
PuBu
Greens
BuGn
GnBu
BuPu
Blues
Lajolla
Turku
Hawaii
Batlow

Diverging

Blue−Red
Blue−Red 2

Conclusions

- Use **HCL-based palettes** (viridis, colorspace) for perceptual uniformity.
- Match palette type to data type: sequential, diverging, or qualitative.
- Always check accessibility with CVD simulations.
- Add redundant encodings (shape, text labels) for critical distinctions.

```r
sessionInfo()
```

```
## R version 4.1.2 (2021-11-01)
## Platform: x86_64-pc-linux-gnu (64-bit)
## Running under: Ubuntu 22.04.5 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.10.0
## LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.10.0
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
##  [3] LC_TIME=en_US.UTF-8        LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=en_US.UTF-8    LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=en_US.UTF-8       LC_NAME=C
##  [9] LC_ADDRESS=C               LC_TELEPHONE=C
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] palmerpenguins_0.1.1 colorspace_2.1-2     patchwork_1.3.2
##  [4] lubridate_1.9.5      forcats_1.0.1        stringr_1.6.0
##  [7] dplyr_1.2.0          purrr_1.2.1          readr_2.2.0
## [10] tidyr_1.3.2          tibble_3.3.1         ggplot2_4.0.2
## [13] tidyverse_2.0.0      conflicted_1.2.0
##
## loaded via a namespace (and not attached):
##  [1] compiler_4.1.2     pillar_1.11.1      RColorBrewer_1.1-3 tools_4.1.2
##  [5] digest_0.6.39      viridisLite_0.4.3  timechange_0.4.0   evaluate_1.0.5
##  [9] memoise_2.0.1      lifecycle_1.0.5    gtable_0.3.6       pkgconfig_2.0.3
## [13] rlang_1.1.7        cli_3.6.5          yaml_2.3.12        xfun_0.56
## [17] fastmap_1.2.0      withr_3.0.2        knitr_1.51         generics_0.1.4
## [21] vctrs_0.7.1        hms_1.1.4          grid_4.1.2         tidyselect_1.2.1
## [25] glue_1.8.0         R6_2.6.1           otel_0.2.0         rmarkdown_2.30
## [29] farver_2.1.2       tzdb_0.5.0         magrittr_2.0.4     scales_1.4.0
## [33] htmltools_0.5.9    labeling_0.4.3     S7_0.2.1           stringi_1.8.7
## [37] cachem_1.1.0
```